

Internet Computing

jQuery

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jQuery

- jQuery is a JavaScript library that makes your JavaScript code writing easier and more powerful
- jQuery code is typically more concise and cleaner than most JavaScript code
- jQuery hides various issues with different browsers from the programmer



Using jQuery

- It is **not necessary** to use jQuery to do things in a web page
- However, it usually helps a lot if you use it
- According to the page below, almost 80% of all websites use jQuery

*[https://w3techs.com/technologies/overview/
javascript_library/all](https://w3techs.com/technologies/overview/javascript_library/all)*

- Let's start by looking at how to include it in your web page

Adding jQuery to Your Page 1/2

- You can use one of the two approaches to add jQuery in your web page
- **Approach 1:**
 - Download jQuery from <http://jquery.com/> and then add a link to the library from your HTML page, i.e.:

```
<script src="jquery-version.js"></script>
```

The version number is typically the latest version that you have downloaded from the jQuery website

Adding jQuery to Your Page 2/2

- **Approach 2:**

- You use the jQuery file from somewhere else
- There are lots of copies of the jQuery library on the web
- Some organisations make a CDN (Content Delivery Network) which means the jQuery library is distributed around the world, and you will automatically receive the file from the closest server

jQuery CDNs

- For example, to link to the Google jQuery CDN, you would do something like this:

<head>

• • •

```
<script src="https://ajax.googleapis.com/ajax/ ➡ line  
➡ libs/jquery/[version]/jquery.min.js"></script>
```

• • •

</head>

Put the latest version number here

- To link to the jquery.com CDN, use this instead:

<head>

• • •

```
<script src="https://code.jquery.com/ ➡ to next line  
➡ jquery-[version].min.js"></script>
```

• • •

</head>

Put the latest version number here

The 'Minified' jQuery

```
<script src="https://code.jquery.com/  
    jquery-[version].min.js"></script>
```



- You are using the 'minimum' version, and sometimes it is called the 'minified' version
- This version uses clever tricks to make the file much smaller, e.g. no unnecessary spaces, variables names that use just 1 or 2 letters, etc
- The 'minimum' version is about 90KB
- The regular version is about 300KB

Basic jQuery Use

- You write code in jQuery similar to what you are doing using DOM functions, i.e.:
 - Access some elements from the DOM and then do something with those elements
 - Trigger some jQuery code by an event
- We will briefly look at the first idea in this presentation and events in another

An Example HTML File 1/2

- We will use this HTML file in the examples in the following slides

```
<!DOCTYPE html>
```

```
<html>
```

```
<head><title>Mac and Cheese</title></head>
```

```
<body>
```

```
<h1 id="name">Mac and Cheese</h1>
```

```
<h2>Ingredients</h2>
```

```
<ul id="ingredients">
```

```
<li>1 box of <a href="...">macaroni</a></li>
```

```
<li>1/4 cup of <a href="...">butter</a></li>
```

```
<li>1/4 cup of <a href="...">flour</a></li>
```

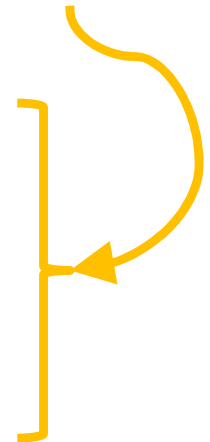
```
<li>1/2 tsp of <a href="...">salt</a></li>
```

```
<li>2 cups of <a href="...">milk</a></li>
```

```
<li>2 cups of <a href="...">cheddar cheese</a></li>
```

```
</ul>
```

*Links are not
shown here to
save space*



- *Continued on the following slide*



An Example HTML File 2/2



- This HTML in a browser is shown on the next slide

```
<h2>Directions</h2>
<ol id="directions">
  <li class="step">Cook the macaroni</li>
  <li class="step">Mix the butter, flour and
    salt in a saucepan</li>
  <li class="step">Add and stir the milk
    until thicken</li>
  <li class="step">Add slowly the cheese
    until fully melt</li>
  <li class="step">Mix with the macaroni</li>
</ol>
</body>
</html>
```



Using class is not just for doing pretty CSS things, it is also used for controlling behaviour

Showing the Example

- This is how the page looks like after loading it in a browser

Mac and Cheese

Ingredients

- 1 box of [macaroni](#)
- 1/4 cup of [butter](#)
- 1/4 cup of [flour](#)
- 1/2 tsp of [salt](#)
- 2 cups of [milk](#)
- 2 cups of [cheddar cheese](#)

Directions

1. Cook the macaroni
2. Mix the butter, flour and salt in a saucepan
3. Add and stir the milk until thicken
4. Add slowly the cheese until fully melt
5. Mix with the macaroni

Everything Starts From \$

- In jQuery, everything that you write starts from the '\$' symbol

- It can be used like a function, e.g.:

```
let myheader = $("h1");
```

- It can also be used to provide some useful functions by writing '\$.', e.g.:

```
if ($.isNumeric("3.3")) ...
```

Selecting Elements in jQuery

- You use these DOM functions to get HTML elements from the DOM:
 - `document.getElementById()`
 - `document.getElementsByTagName()`
- In jQuery, you use the `$(...)` function to select elements using *CSS selectors*
- Let's see a few examples in the next slides

Using the id Attribute

- Let's select the `<h1>...</h1>` on the right using its `id`

```
<h1 id="name">  
    Mac and Cheese  
</h1>
```

- Using DOM functions:

```
let name =  
    document.getElementById("name");
```

- In jQuery:

```
let name = $("#name");
```

*A CSS selector
referring to the id
called 'name'*

The jQuery Object

- Important notes about `$(...)`:
 - The result returned by `$(...)` is a *jQuery object*
 - It is **not** returned as a DOM element
- You **cannot** use the jQuery object like what you have done using the DOM element
- An example with `innerHTML` is shown in the next slide

Using the jQuery Object

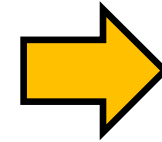
- It does not work this way using the jQuery result:

```
let name = $("#name");  
alert(name.innerHTML);
```

*innerHTML not working
for the jQuery object*



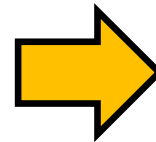
```
<h1 id="name">  
  Mac and Cheese  
</h1>
```



This page says
undefined

- In jQuery, you do it like this:

```
let name = $("#name");  
alert(name.html());
```



This page says
Mac and Cheese

Selecting Elements By Tag Name

- Previously, you select elements of the same HTML tag using this code, e.g.:

```
...  
<h2>Ingredients</h2>  
...  
<h2>Directions</h2>  
...
```

```
let headers =  
    document.getElementsByTagName("h2");
```

- In jQuery, you do that by, e.g.:

```
let headers = $("#h2");
```

The CSS selector referring to all <h2>

Having Multiple Elements

- Most of the operations under the jQuery object can work with multiple elements
- You can read the number of elements in a jQuery object using its `length` property, e.g.:

```
$("#title").length // return 1
```

```
<title>Mac and Cheese</title>
```

```
$("#h2").length // return 2
```

```
<h2>Ingredients</h2>
```

```
<h2>Directions</h2>
```

Selecting Elements By Class

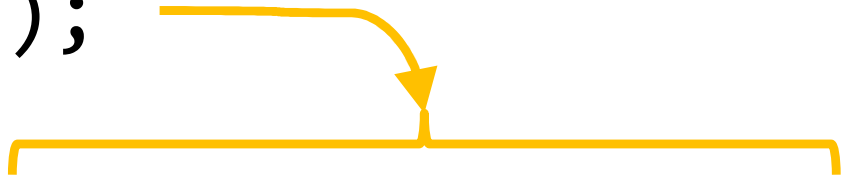
- You can also select elements using their class names, i.e. class attribute

- Here is an example:

Selecting five s

```
let steps = $(".step");
```

- In the example HTML, no visual style has been applied to the class

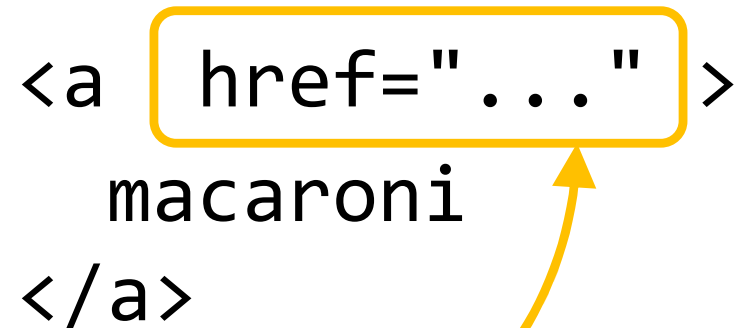


```
<li class="step">Cook...  
<li class="step">Mix ...  
<li class="step">Add ...  
<li class="step">Add ...  
<li class="step">Mix ...
```

Reading Attributes

- Once you get the elements, you can start to do different things with them
- If you want to read the attribute of an element, simply do this, e.g.:

`<a href="..." macaroni >`
`</a>`

A diagram illustrating how to read an attribute from an HTML element using jQuery. It shows an HTML anchor tag: `<a href="..." macaroni >` followed by `</a>`. The attribute `href="..."` is enclosed in a yellow rounded rectangle. A yellow curved arrow points from the `href` property in the jQuery selector `$("#a").attr("href");` to the highlighted attribute in the HTML tag.

```
let url = $("#a").attr("href");
```

- For a jQuery object with multiple elements, jQuery always read **only** the attribute of the first element

Changing Attribute Content

- You can change the attributes of **all** elements in the jQuery object, like this:

```
$("a").attr("href",  
            "https://en.wikipedia.org")
```



Every href is changed

```
<a href="https://en.wikipedia.org">macaroni</a>  
<a href="https://en.wikipedia.org">butter</a>  
<a href="https://en.wikipedia.org">flour</a>  
<a href="https://en.wikipedia.org">salt</a>  
<a href="https://en.wikipedia.org">milk</a>  
<a href="https://en.wikipedia.org">  
cheddar cheese</a>
```

Reading CSS Properties

- You can read and change CSS properties using `.css()` (not the `style` attribute!)
- Similar to `.attr()`, you can read CSS properties of the **first** element only, e.g.:

```
let color = $("li.step").css("color");
```

- The above code reads the `color` property of the first `<li>` with the class name `step`

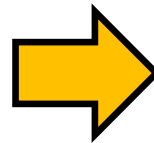
Changing CSS Properties

- You can change the CSS property of **all** elements returned by jQuery, e.g.:

```
$("li.step").css("color", "red");
```

- The above code changes all matching `<li>` to red, like this:

1. Cook the macaroni
2. Mix the butter, flour and salt in a saucepan
3. Add and stir the milk until thicken
4. Add slowly the cheese until fully melt
5. Mix with the macaroni



1. Cook the macaroni
2. Mix the butter, flour and salt in a saucepan
3. Add and stir the milk until thicken
4. Add slowly the cheese until fully melt
5. Mix with the macaroni

Reading the Element Content

- Using the DOM, you read the 'inner content' of an element using `innerHTML`
- In jQuery, you can do that using `.html()` and `.text()`
 - `.html()` can read, also can create HTML tags
 - `.text()` can read, also can create plain text content

Using .html()

- Using .html() is like innerHTML, e.g.:

```
$("#name").html("Yummy Mac and Cheese");
```

Yummy Mac and Cheese

- .html() works for HTML content too, i.e.:

```
$("#name").html("<i>Yummy</i> Mac and Cheese");
```



Yummy Mac and Cheese

Using .text()

- Using .text() gives you a different result when the content has HTML, e.g.:

```
$("#name").text("<i>Yummy</i> Mac and Cheese");
```



<i>Yummy</i> Mac and Cheese

- You would want to use .text() sometimes when you want to show special characters, i.e. <, >, & and so on, as plain text

jQuery and DOM

- In some cases, you may want to get the DOM elements from the jQuery object
- You can convert between a jQuery object and a DOM element easily, e.g.:

- From a jQuery object to a DOM element

```
let domHeader = $("h2")[0];  
domHeader.innerHTML = ...;
```

 You can get the element you want, e.g. [1], [2]

- From a DOM element to a jQuery object

```
let jqHeader = $(domHeader);  
jqHeader.html(...);
```